

provided with Google Cardboard viewers, materials and related training, so that they can take virtual educational tours. Google has developed over 100 exciting journeys, from outer space to museums and the world under the ocean. Once a school decides on a schedule for the virtual expeditions, Google volunteers will arrive at the venue to help the teachers and ensure that children have a smooth and enjoyable experience without any glitches.

We are seeing several innovative applications closer home, too. Common Floor, a real-estate company, has developed a VR app available on Google Play, which lets users visit properties virtually without moving out of their home. Common Floor also sells a Cardboard based headgear called Retina. The views are as if you are at the venue itself, but save you the hassle of travelling here and there to view multiple properties.

Imaginate is developing VR content and intends to be a repository for educational content. "I am personally excited about using VR as a virtual training or experiential medium for several use cases, such as training—be it in manufacturing, automobile, psychiatry, physiotherapy, education of all sorts, etc.," says Satyanarayana. There is a growing market for VR in India, and Imagine has been developing cutting-edge solutions for several B2B enterprises in fields like marketing, engineering, automation, training and defence.

Way back in 2004, Satyanarayana developed an AR based liver surgery apparatus using a device called LiEye (like Google Glass), which enabled a surgeon performing open-liver surgeries to see through a liver and identify the tumours hidden inside.

In 2012, he was lauded as one of the Innovators under 35 by MIT Technology Review for an AR-enabled virtual trial room called

TrialAR, now rechristened as Dressy.

Dressy is a plug-and-play fitting room for offline and online e-commerce stores and mobile apps. It helps buyers to instantly try out apparel as well as accessories, and share their shopping experience with friends and family members located elsewhere. Its digital catalogue and analytics engine serve dual purposes of helping customers find what they want and help-



Google Jump assembles 360 videos that allow users to experience a scene in every direction

ing retailers understand customer preferences better.

Hash Media has also recently developed a similar solution called Virtual Mirror, which uses a smartphone or computer's camera to help users try out stuff before they buy and to share the images with friends on social media to seek their suggestions. Virtual Mirror can be used by individuals or installed as kiosks in shops and public places.

Hash Media also has a VR solution that lets manufacturing companies and civil engineering companies to construct a plant or building virtually for training and maintenance purposes. "We are also developing a virtual shopping mall, where you enjoy an experience like an actual shopping mall. You can roam in the mall and purchase products, which will be delivered to your home. You just need VR glasses," says Lalpura.

Another very interesting product from Hash Media helps people overcome stage fright and improve their public speaking skills. Using the VR tool, users can give speeches in front of a virtual crowd. They can rehearse as many times as they want, till they get the speech right. The crowd is even programmed to ask certain pre-determined questions, providing a more realistic experience.

Exciting times

We are living in an exciting time for VR. Whether it is a basic low-cost headgear you seek to start dapping with VR or an advanced one for a completely immersive experience, you are sure to get what you seek, with the added benefit of being able to turn your room into a 3D space!

Even more exciting is the extensive content available today, as well as the platforms to make your own. So there is no doubt that the near future is going to be very exciting.

But how would VR be, well into the future? Would it be possible to create immersive environments without using glasses? Imagine a school practice session where multiple kids can engage in the same virtual environment, practicing with others, hopefully without glasses. Perhaps, that too will be real one day.

"VR will revolutionise the way we live, play and learn. I will say that the technology is still in a child state. Today's VR glasses and other hardware are just foot soldiers. In the coming years, you will see a big paradigm shift, even in the way we compute using the smartphone.

"Hologram technology is also a front-runner. Now, we are able to create holograms in air using lasers. So in the future we would not need a headgear to display graphics for VR. It will be created directly in air like we have seen in science fiction films like Star Wars," signs off Lalpura. **EFY**